🗺️ GEOSPATIAL & DIGITAL HUMANITIES ARCHITECTURE

Bible Map Crafter & Thematic Spatial Studio

Adapting modern, client-side geospatial engines for ancient history, literature, archaeology, and educational atlases without anachronistic modern road grids.

📜 Dual-Plate

Instant split-peel slider between Ancient Parchment & Modern Satellite.

📍 1,200+ Sites

Authoritative gazetteers from OpenBible, Pleiades, and Wikidata.

🎯 Hitbox Engine

8px bounding-box click tolerance on travel routes and ancient empires.

⚡ Serverless

MapLibre GL JS 4.x hosted on Google Cloud Storage for pennies a month.

Ancient Historical Atlas Template

Strips away modern 4-lane motorways and urban grids in favor of vintage parchment topography.

📜 Vintage Parchment Shaded Relief

Replaces modern asphalt tiles with bespoke hillshading, warm sepia relief, and bathymetric tints to reflect ancient terrain.

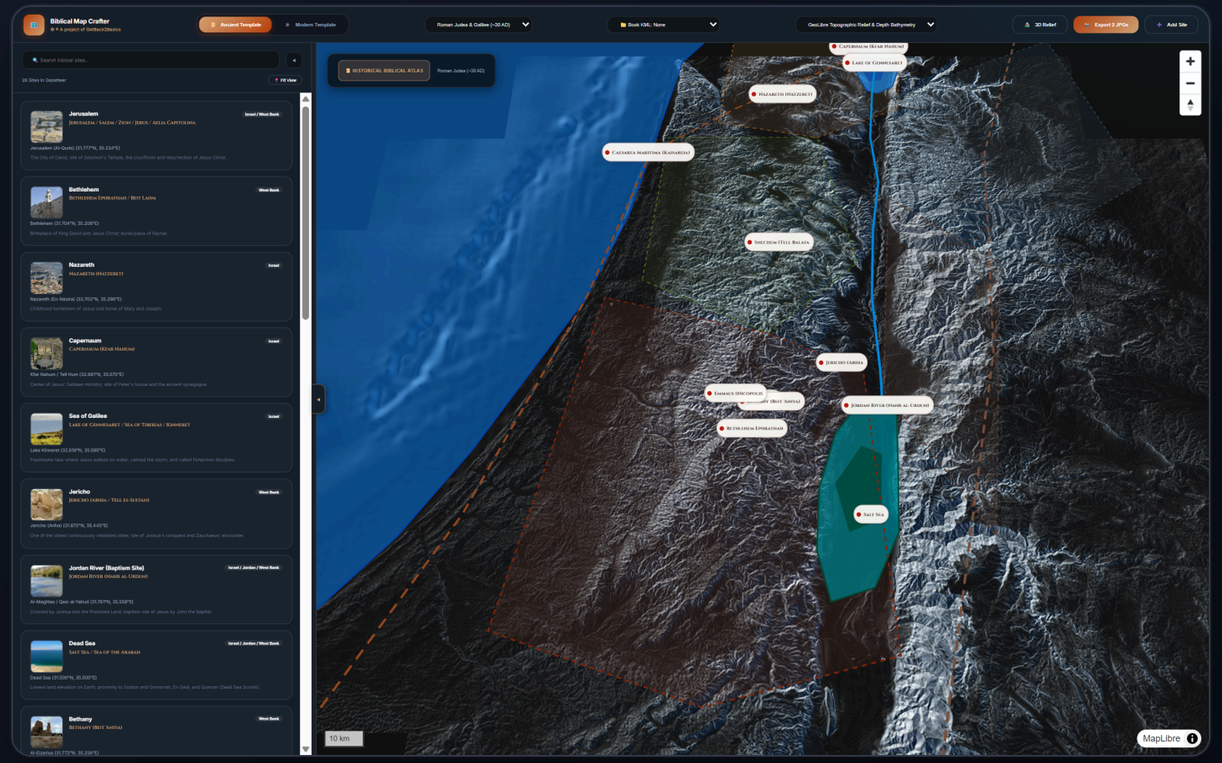
⏳ Chronological Era Filtering

Instant client-side filtering for 5 major biblical epochs: Patriarchal (~2000 BC), Exodus, United Monarchy, Roman Judea (~30 AD), and Early Church (~50-95 AD).

📁 Dynamic KML Vector Overlays

Streams authoritative chapter routes and territorial boundaries on demand via `@tmcw/togeojson` with automatic bounding-box fitting.

MapLibre GL JS 4.x | 3D DEM Terrain | OpenTopoMap Shading | GeoJSON / KML



Site Popups with Photo Carousels & Lightbox

Connecting ancient coordinates to verified archaeological excavations and peer-reviewed media.

🏛️ Tri-Gazetteer Cross-Referencing

Synchronizes over 1,200 ancient Near Eastern locations against **OpenBible.info**, **Pleiades**, and **Wikidata** to eliminate duplicate and disputed coordinates.

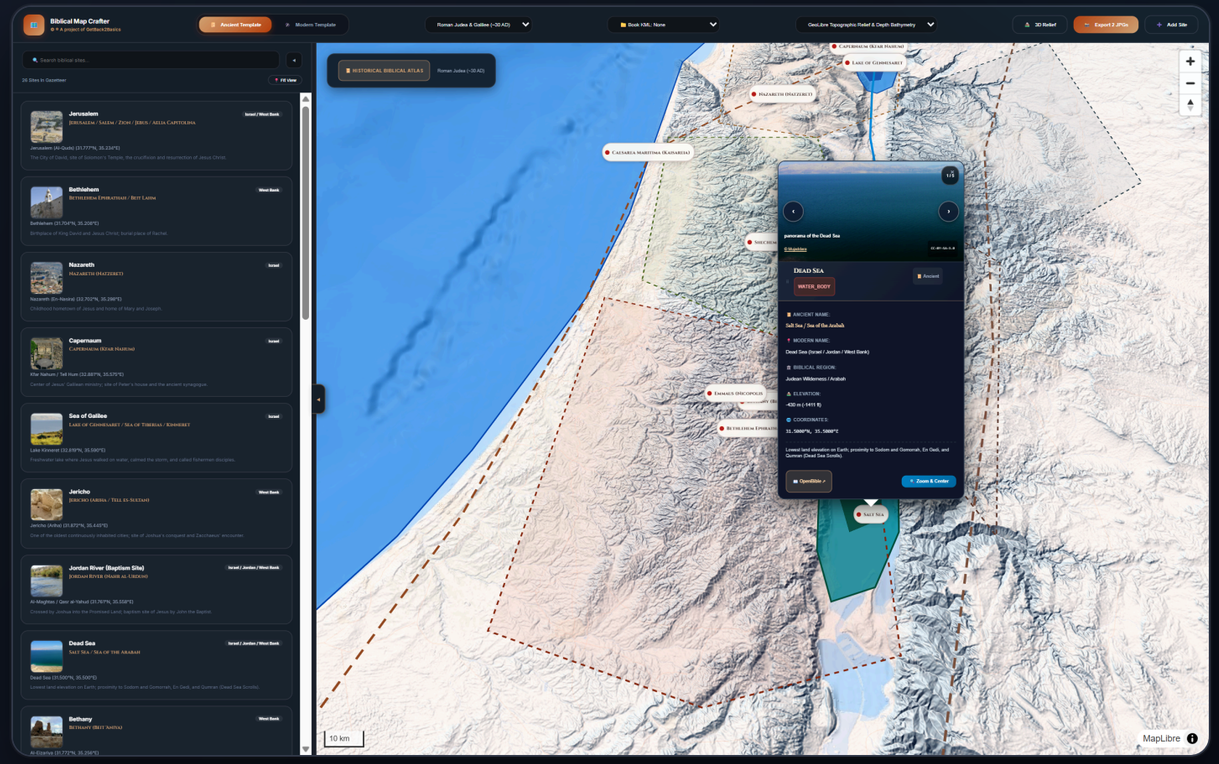
🖼️ Responsive In-Map Photo Carousels

Every prominent site features a multi-image drawer carousel with CC-licensed photography, photographer credit, and Wikimedia Commons source badges.

🔍 Fullscreen Lightbox & In-Browser Editor

Keyboard-navigable modal lightbox for high-resolution artifact inspection, plus an in-browser site editor to update coordinates and descriptions.

Wikimedia Commons API | Pleiades Gazetteer | Modal Lightbox | CC Attribution



Modern Satellite Template Integration

One-click switch to high-resolution satellite imagery while maintaining historical vector markers.

🛰️ Seamless Imagery Switching

Instantly shifts the basemap to ESRI World Imagery without dropping active site selections, travel vectors, or historical search state.

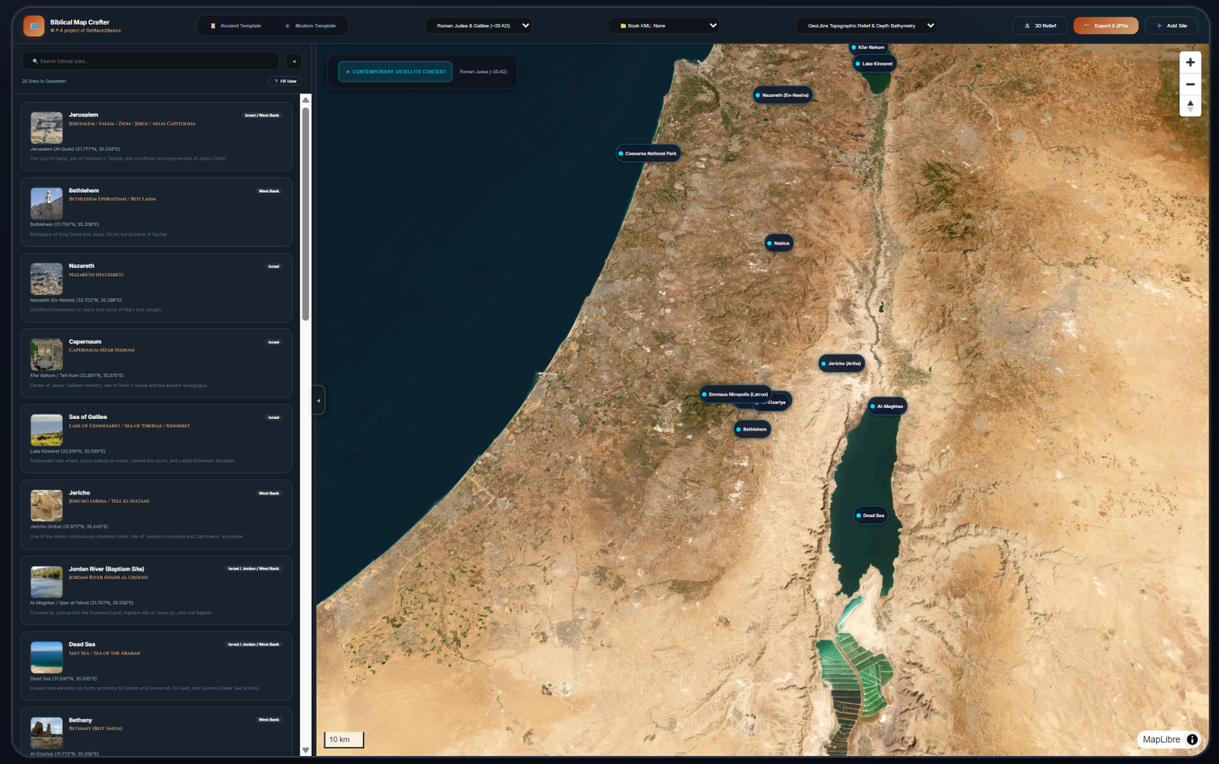
⛰️ Real-World Topography & Tell Verification

Enables researchers, students, and educators to correlate ancient texts with geological river bends, wadis, mountain passes, and archaeological tells.

📸 High-Resolution Dual-Plate Exporter

Direct WebGL canvas capture that exports synchronized Ancient and Modern JPEG plates with matching bounds for publication and offline printing.

ESRI World Imagery | Canvas Capture API | Synchronized Bounds | High-DPI Export



Dual-Plate Split Peel Slider (React)

Consuming generated plates in modern web applications to provide real-time interactive exploration.

🪫 Interactive Real-Time Peel Slider

Users drag an intuitive split slider to peel back the ancient historical plate and reveal modern satellite photography underneath in real time.

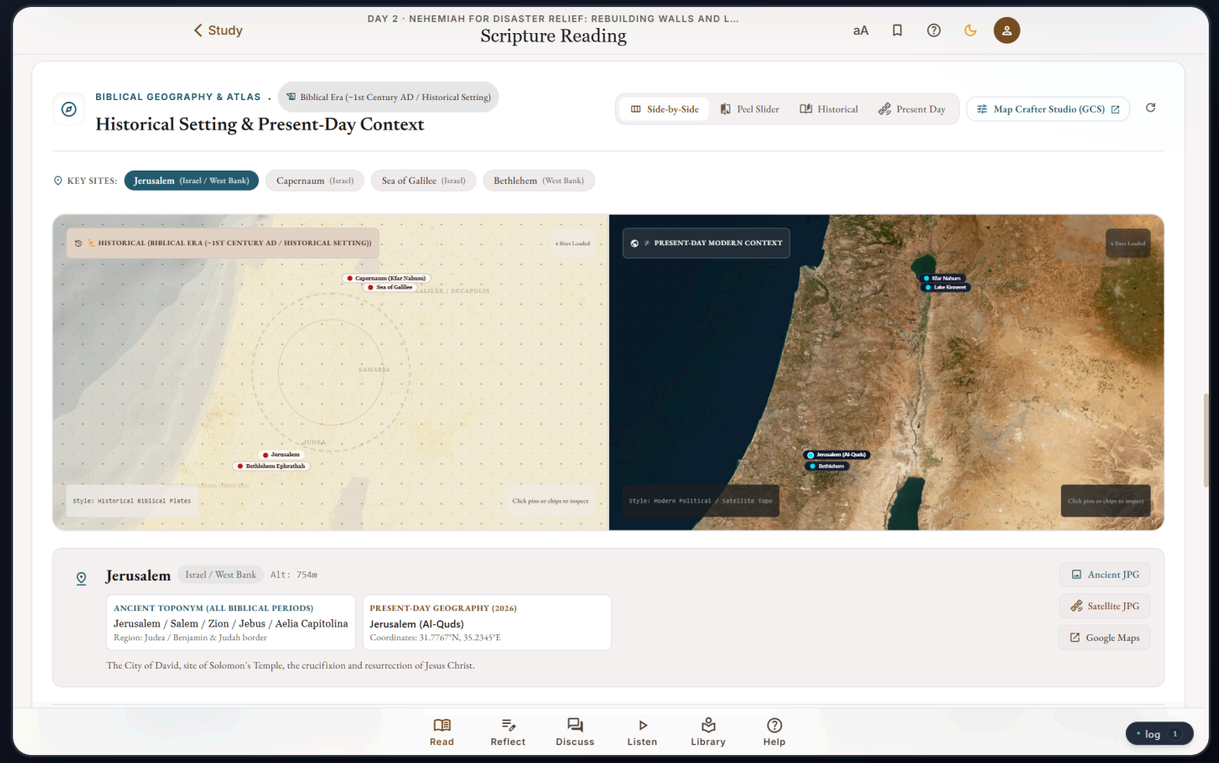
🧭 In-Context Geographic Discovery

Seamlessly embedded into curriculum lessons, displaying historical plate context, archaeological significance, and direct GCS studio deep-links.

🔗 Deep-Linked Studio Handoff

A single click on "Map Crafter Studio (GCS)" opens the standalone studio centered on the exact coordinates and era of the active lesson.

React 18 / TypeScript | Touch & Mouse Drag | Deep Linking | CSS Clip-Path



Live App: **biblestudy-crafter-uc2dhzwjsa-uc.a.run.app**

The Upstream Engine: Highlight Map Crafter

From short-term rental concierge and travel isochrones to universal thematic cartography.

🏠 Property Hub & Travel-Time Isochrones

Initially created for our Lake Macquarie guest stay (21 Bournville Rd, Rathmines) using Valhalla to calculate true 5/10/15/30-minute drive & walk polygons.

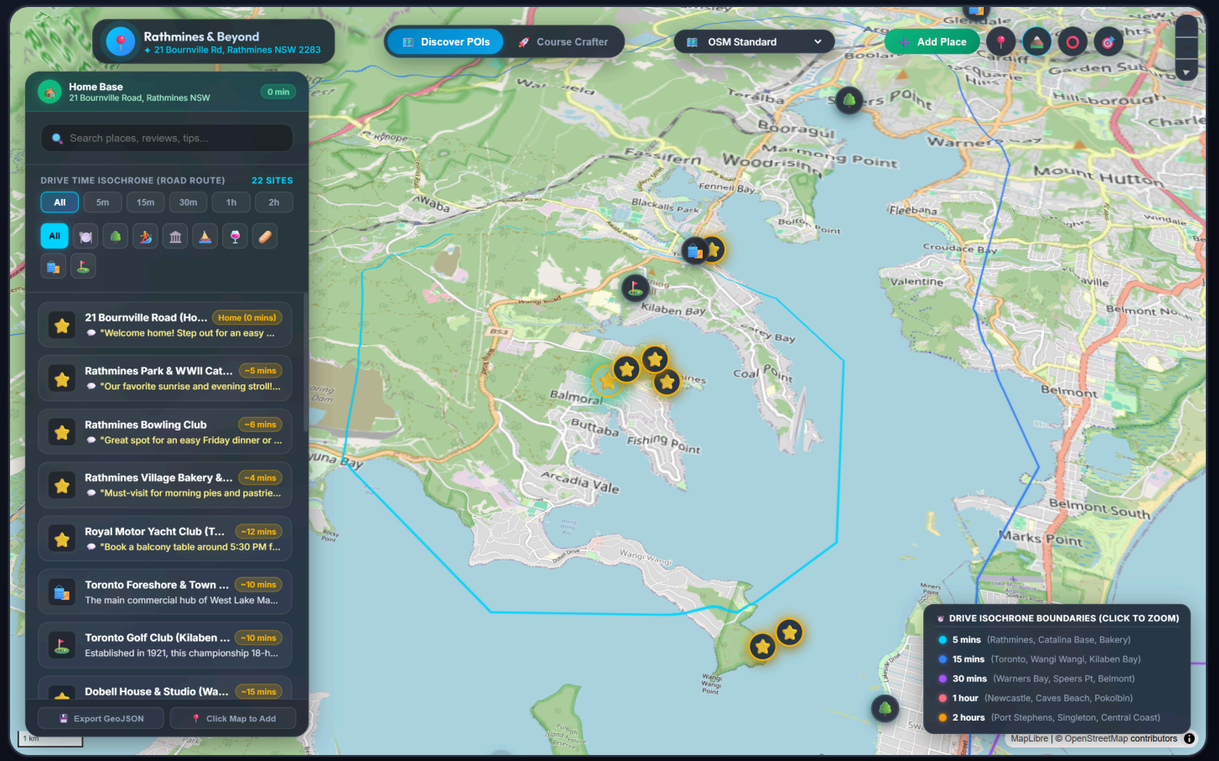
🔄 Contributing Upstream Innovations

Enhancements engineered in Bible Map Crafter (8px vector click hitboxes, multi-photo carousels, dynamic KML vector streaming) were contributed straight back upstream.

🤖 AI Welcome Guide with Google NotebookLM

Integrated conversational AI concierge transforming property guides, house manuals, and local secrets into a 24/7 guest assistant.

Valhalla Isochrones | Turf.js Buffering | NotebookLM Integration | OpenStreetMap



Applications Far Beyond Biblical Studies

A reusable, zero-backend architecture for digital humanities, education, and cultural heritage.

🏛️ Classical History & Literature

Interactive companions tracing Homer's *Odyssey*, the campaigns of Alexander the Great, Silk Road trade caravans, or Roman highways across Britannia without modern border distractions.

🎓 K-12 & Higher EdTech

Engaging classroom modules allowing students to interactively peel between ancient empires (Mesopotamia, Pharaonic Egypt, the Indus Valley) and present-day sovereign nations.

🏺 Museums & Cultural Heritage

Lightweight, embeddable touchscreen kiosks displaying artifact provenance, excavation strata, and historical context without heavy, expensive desktop GIS licenses.

🌿 Environmental & Historical Tracking

Visualizing changing river deltas, historical deforestation, ancient maritime routes, or indigenous territorial boundaries with high-resolution satellite overlay.

🚀 OPEN SOURCE & FREE TO ADAPT

Explore the Live Tools & Source Code

Both repositories are completely open-source under MIT & ISC licenses. Feel free to fork, plug in your own historical GeoJSON/KML data, or adapt the spatial engine for your institution.

🌐 Live Map Studio

storage.googleapis.com/biblestudy-map-crafter

📖 Live App (Slider)

biblestudy-crafter-uc2dhzwjsa-uc.a.run.app

💻 Bible Map Crafter

github.com/GetBack2Basics/BibleStudy-Crafter

🗺️ Upstream Engine

github.com/GetBack2Basics/Highlight_Map_Crafter